

Testability and Observability

How to design and investigate systems that provide useful evidence for QA.

What is testability?

Testability is the degree to which a system allows criteria to be established, conditions to be controlled, results to be observed and behavior to be investigated. Useful logs, accessible APIs, correlation IDs and controllable data reduce testing cost.

Observability for QA

Logs, metrics and traces help answer what happened, where and when. In automation, screenshots, videos, Playwright traces, console and Network data make failures easier to diagnose.

Controllability

QA needs ways to create data, change states, simulate dependencies and reproduce scenarios. Mocks, stubs, feature flags and service virtualization can improve controllability.

Practical assessment

Ask: can I prepare the state, observe the outcome, identify the version, correlate requests and reproduce failures? The more answers are no, the higher the testing risk and cost.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.